

MLPerf(TM) Storage v3.0 - System Description

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System name: aws_m7i48xlarge_i3en12xlarge_32c_32s_512p
Workload: Checkpointing - llama3-405b, 512 MPI processes (CLOSED, 16/client)
Storage system: Alluxio Enterprise AI 3.9 (RC18) over AWS S3, via Alluxio-FUSE (POSIX)
Result: Save 147.0 GiB/s / Load 124.3 GiB/s (10 checkpoints; two-invocation split)
Cloud: AWS (us-west-1), single availability zone

1. Hardware configuration & node counts

Clients (host nodes): 32 x m7i.48xlarge

- 32 FUSE checkpoint clients, 16 MPI processes each (512 total)
- CPU: 192 vCPU (Intel Xeon Platinum 8488C, Sapphire Rapids) Memory: 768 GiB
- Storage: EBS gp3 (root) Network: 50 Gbps ENA

Storage nodes: 32 x i3en.12xlarge

- 32 Alluxio storage workers (NVMe RAID0 page store)
- CPU: 48 vCPU (Intel Xeon Platinum 8259CL) Memory: 384 GiB
- Storage: 4x 7.5 TB NVMe SSD -> RAID0 (~28 TB) page store Network: 50 Gbps ENA
- Deployment: EKS (K8s 1.31) via Alluxio operator 3.5.1 + bundled FoundationDB operator
- Dedicated FoundationDB metadata tier: 3x fdb-log (m5.2xlarge, tlogs + coordinators) + 3x fdb-storage/cluster_controller/stateless (m5.2xlarge); redundancy=double;
- coordinatorSelection=log; tlogs on disks SEPARATE from the worker page store.
- Coordinator + etcd: 1x m5.2xlarge. Client attach: CSI FUSE (mountPropagation HostToContainer).

2. System & network topology

Benchmark access is POSIX/file via Alluxio-FUSE; AWS S3 is the under-file-system (UFS). All nodes in one AWS us-west-1 availability zone over 50/100 Gbps ENA.

```
32x m7i.48xlarge --CSI FUSE (POSIX, write)--> 32x i3en.12xlarge workers (NVMe RAID0)
(MPI, 16 procs ea, 512) write-back cache | metadata commits
                                         DEDICATED FDB TIER: 3x log (tlogs, coordinators)
                                         + 3x storage/cc/stateless
                                         coordinator + etcd (1 node)
                                         |
                                         AWS S3 (UFS, TRANSIENT ~0)
```

3. Software stack & versions

- Alluxio Enterprise AI 3.9 (RC18) (distributed cache + write-back)
- Alluxio-FUSE (POSIX client)
- FoundationDB 7.3.63 (write-cache metadata)
- etcd 3.5.x (mount table + worker membership)
- Ubuntu 24.04 LTS
- OpenJDK 17
- mlpstorage 3.0.x (uv-managed) + DLIO 3.0.x
- Backing store: AWS S3 (UFS)

4. Non-default configuration options

- Client write cache: alluxio.write.cache.enabled=true + FoundationDB (FdbBasedFileSystem)
- Checkpoint path routed to policyMode: TRANSIENT (write-back cache; S3 persist ~0 during run)
- Multipart write buffer: part.size=64MiB, buffer.number=512, threads=512, max.buffers.per.stream=32
- alluxio.dora.client.ufs.fallback.enabled=true; list.status.from.ufs.enabled=false
- worker.network.netty.file.transfer=TRANSFER; frame.decoder.buffer.cumulator=COMPOSITE
- FUSE JVM: -Xmx96g -XX:MaxDirectMemorySize=48g
- Two-invocation split: write (10/0) -> drop_caches -> read (0/10); inter-phase gap <=30s (rules 4.7.1)

5. Notes

Simulated accelerators (no GPU). Machine-readable system parameters are in the companion `systemname.yaml`.
CLOSED workload parameters per `mlpstorage`; storage-system configuration is unconstrained in CLOSED. Dataset
backing store: AWS S3.